11 Waste Management Implications

11.1 Introduction

11.1.1.1 This section identifies the quality and quantity of waste generated from construction and operation phases of the Project, and evaluates the potential waste management implications that may result from waste generated during these phases.

11.1.1.2 Mitigation measures and good site practices, including waste handling, storage and disposal, have been recommended with reference to relevant waste legislation and management guidelines.

11.1.1.3 The waste management implications have been assessed in accordance with the requirements of Annexes 7 and 15 of the Technical Memorandum on Environmental Impact Assessment Ordinance (EIAO-TM) as well as requirements set out under Clause 3.4.8 of the EIA Study Brief.

11.2 Environmental Legislation, Standards and Guidelines

11.2.1.1 The relevant legislation and associated guidance notes relate to the study for the assessment of waste management implications include:

- Waste Disposal Ordinance (WDO) (Cap. 354) and subsidiary Regulations.
- Environmental Impact Assessment Ordinance (EIA Ordinance) (Cap. 499), EIAO-TM Annex 7 and Annex 15.
- Land (Miscellaneous Provisions) Ordinance (Cap. 28).
- Public Health and Municipal Service Ordinance (Cap. 132) - Public Cleansing and Prevention of Nuisances Regulation.

11.2.1.2 Under the Waste Disposal Ordinance, some of the regulations are relevant to EIA, including:

- Waste Disposal (Chemical Waste) (General) Regulation (Cap. 354).
- Waste Disposal (Charges for Disposal of Construction Waste) Regulation (Cap. 354).

11.2.2 Waste Disposal Ordinance (WDO)

11.2.2.1 The Waste Disposal Ordinance (WDO) prohibits any unauthorised disposal of wastes. Construction waste, defined under Cap. 354N of the WDO, refers to a substance, matter or thing which is generated from construction works. It includes all abandoned materials, whether processed or stockpiled or not, before being abandoned, but does not include sludge, screenings or matter removed or generated from desludging, desilting or dredging works.

11.2.2.2 Under the WDO, wastes can only be disposed of at designated waste disposal facilities licensed by Environmental Protection Department (EPD). Breach of this Ordinance can lead to a fine and/or imprisonment. The WDO also stipulates the requirements for issuing licenses for the collection and transportation of wastes.

11.2.3 Waste Disposal (Charges for Disposal of Construction Waste) Regulation

11.2.3.1 Under the Waste Disposal (Charges for Disposal of Construction Waste) Regulation, construction waste delivered to a landfill for disposal must not contain more than 50% by weight of inert material. Construction waste delivered to a sorting facility for disposal must contain more than 50% by weight of inert material, and construction waste delivered to a Public Fill Reception Facilities for disposal must consist entirely of inert material.

11.2.4 Waste Disposal (Chemical Waste) (General) Regulation

11.2.4.1 Under the WDO, the Chemical Waste (General) Regulation provides regulations for chemical waste control, and administers the possession, storage, collection, transport and disposal of chemical wastes. EPD has also issued a 'guideline' document, the Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes (1992), which details how the Contractor should comply with the regulations on chemical wastes.

11.2.5 Land (Miscellaneous Provisions) Ordinance

11.2.5.1 The inert portion of C&D materials may be taken to public filling facilities including public filling area, public filling barging points and stockpiling areas. These facilities usually form part of land reclamation schemes and are operated by CEDD. The ordinance requires Dumping Licenses (to be issued by CEDD) to be obtained by individuals or companies, who deliver inert C&D materials to the public filling facilities.

11.2.6 Public Cleansing and Prevention of Nuisances Regulation

11.2.6.1 This regulation provides control on illegal tipping of wastes on unauthorised (unlicensed) sites.

11.2.7 Construction & Demolition Material Management Plan (C&DMMP)

11.2.7.1 Measures have been introduced under Section 4.1.3 of the Project Administrative Handbook for Civil Engineering Works (2014 Edition) published by CEDD to enhance the management of C&D materials and to minimise their generation at source. The enhancement measures include (i) drawing up a Construction and Demolition Material Management Plan (C&DMMP) at an early design stage to minimise C&D materials generation and encourage proper management of such materials; (ii) vetting of the C&DMMP prior to upgrading of the project to Category A in the Public Works Programme; and (iii) providing the Contractor with information from the C&DMMP in order to facilitate the preparation of the Waste Management Plan (WMP) and to minimise C&D materials generation during construction. Projects generating less than 50,000m^3 C&D materials or importing less than 50,000m^3 of fill material are exempted from the C&DMMP. The comments on the C&DMMP given by the Public Fill Committee is attached in **Appendix 11.1**, and will be incorporated at the detailed design and construction stages.

11.2.7.2 ETWB TCW No. 19/2005, Environmental Management on Construction Site, sets out the policy, procedures and requirements for contractor to prepare and implement an Environmental Management Plan for on-site sorting and waste reduction of C&D materials.

11.2.8 Other Relevant Guidelines

11.2.8.1 The documents and guidelines in the following table also relate to waste management and disposal:

Table 11.1 Other relevant documents and information

Bureau / Department	Documents / Guidelines / Technical Circulars
Development Bureau	WBTC No. 2/93, Public Dumps WBTC No 2/93B, Public Filling Facilities WBTC No. 16/96, Wet Soil in Public Dumps WBTC Nos. 4/98 and 4/98A, Use of Public Fill in Reclamation and Earth Filling Project WBTC No. 12/2000, Fill Management WBTC No. 19/2001, Metallic Site Hoardings and Signboards WBTC No. 12/2002, Specification Facilitating the Use of Recycled Aggregates ETWB TCW No. 34/2002, Management of Dredged / Excavated Sediment ETWB TCW No. 19/2005, Environmental Management on Construction Site DEVB TCW No. 06/2010, Trip-ticket System for Disposal of Construction and Demolition Material DEVB TCW No. 08/2010, Enhanced Specification for Site Cleanliness and Tidiness DEVB TCW No. 02/2011, Encouraging Use of Recycled and other Green Materials in Public Works Projects DEVB TCW No. 09/2011, Enhanced Control Measures for Management of Public Fill
CEDD	Project Administrative Handbook for Civil Engineering Works, Section 4.1.3 of Chapter 4, 2016 Edition CEDD TC No. 03/2015, Management of Construction and Demolition Materials
EPD / CEDD	New Disposal Arrangements for Construction Waste (1992)
EPD	Waste Disposal Plan for Hong Kong (December 1989) ProPECC PN2/97, Handling of Asbestos Containing Materials in Buildings Waste Reduction Framework Plan, 1998 to 2008 A Policy Framework for Management of Municipal Solid Waste (2005 -2014), (December 2005) Code of Practice on the Packaging, Labeling and Storage of Chemical Wastes (1992) Practice Guide for Investigation and Remediation of Contaminated Land (2011) Code of Practice on the Handling, Transportation and Disposal of Asbestos Waste Guidance Note for Contaminated Land Assessment and Remediation, (2007) Guidance Manual for Use of Risk-based Remediation Goals for Contaminated Land Management, (2007)

Bureau / Department	Documents / Guidelines / Technical Circulars
PlanD	Hong Kong Planning Standards and Guidelines, Chapter 9 (Section 6 – Waste Management)

11.3 Assessment Methodology

11.3.1.1 The assessment of waste management implications from handling, storage, collection, transportation and disposal of solid waste materials generated by the landuse proposals have been undertaken in accordance with Annexes 7 and 15 of the EIAO-TM.

11.3.1.2 The waste management hierarchy has been applied in the assessment and development of mitigation measures for waste. The waste management hierarchy is a concept which shows the desirability of various waste management methods and comprises the following in order of preference:

- avoidance;
- minimisation;
- recycling/reuse;
- treatment; and
- disposal.

11.3.1.3 All opportunities for reducing waste generation have been assessed based upon the following factors:

- avoiding or minimising waste generation throughout design, construction and operation phases;
- adopting better management practices to promote segregation materials;
- reuse and recycling on site or other projects; and
- diverting C&D materials to public fills as far as possible.

11.3.2 Analysis of Activities and Waste Generation

11.3.2.1 The quantity, quality and timing of the waste arising as a result of the construction and operation activities of the Project site have been estimated, based on the sequence and duration of these activities. The design, general layout, construction methods and programme to minimise the generation of public fill/inert C&D materials for other construction works have been considered.

11.3.2.2 The potential waste management implications associated with the handling, transportation and disposal of waste arising from the construction works have been assessed with reference to the following approach:

- estimation of the types, timing and quantities of the wastes to be generated and fill to be imported; and
- assessment of the potential waste management implications on the capacity of collection, transfer and disposal facilities.

11.3.2.3 The waste generation and disposal rate adopted in the assessment is based on "*Monitoring of Solid Waste in Hong Kong 2015*".

11.3.3 Proposal for Waste Management

11.3.3.1 Prior to considering the disposal options for various types of wastes, opportunities for reducing waste generated, on-site or off-site re-use and recycling have been evaluated. Measures which can be taken in the planning and design phases (e.g. by modifying the design approach) and in the construction phase for maximizing waste reduction have been separately considered.

11.3.3.2 After considering all the opportunities for reducing waste generation and maximizing re-use, the types and quantities of the remaining wastes required to be disposed of have been estimated and the disposal options for each type of wastes have been discussed. The disposal method recommended for each type of waste has taken into account of the assessment result.

11.3.3.3 The impacts caused by handling (including labelling, packaging and storage), collection, and reuse/disposal of wastes have been addressed and appropriate mitigation measures have been proposed.

11.4 Identification and Evaluation of Waste Management Implications

11.4.1 Construction Phase

11.4.1.1 As mentioned in **Chapter 1**, the Project site comprises of Potential Development Area (PDA) as well as supporting infrastructure / development works.

11.4.1.2 The following activities, including the DP and non-DP elements in the RODP (as mentioned in **Sections 1.5** and **1.6**), have been included in the waste management implication assessment for the construction phase:

- Construction of Trunk Roads (TSWW Interchange) (DP1);
- Construction of new Primary Distributor Roads (TYST Interchange) (DP2);
- Construction of two new Distributor Roads (Road D1 to Road D2) (DP3);
- Environmentally Friendly Transport Services (EFTS) (DP4) (by others);
- Construction of slip roads at the TYST Interchange (DP5);
- Construction of partly depressed road/ underpass located at TSWW Interchange and full enclosures at TYST Interchange (DP6);
- Construction of new YLS STW (Site OU3.3) (DP7) (by others);
- Construction of two new SPSs (Sites OU1.12 and OU2.2) (DP8);
- Construction of Reclaimed Water Service Reservoir for reuse of reclaimed water (DP9);
- Yuen Long Nullah revitalisation/decking along Kung Um Road and Kiu Hing Road (DP10) (by others);
- Construction of Residential Uses, Storage and Workshop Uses, Government, Institution or Community, Education, Public Utilities, Road and River Channels, Amenity, Open Space; and
- Other miscellaneous construction works.

11.4.1.3 The waste implication during construction phase from the Project site has been assessed and presented in the following sub-sections.

11.4.1.4 The waste implication during construction due to the development of the Project site has been assessed and presented in the following sub-sections. The main activities which would potentially result in the generation of waste include:

- site clearance;
- site formation;
- construction of road networks; and
- construction of the new buildings and infrastructures.

11.4.1.5 A variety of type of wastes would be generated during construction phase that can be divided into the following distinct categories based on their compositions:

- site clearance wastes;
- site formation wastes;
- building construction wastes;
- excavated sediment;
- contaminated soil;
- asbestos containing materials
- chemical wastes; and
- general refuse.

11.4.1.6 The construction activities of the Project consist of site clearance, site formation and construction of new buildings and infrastructures.

11.4.1.7 The estimated amount of different types of waste to be generated by different construction activities is summarised in **Table 11.2**.

Table 11.2: Estimated amount of different type of wastes to be generated by different construction activities

Activities	Quantity of Waste to be Generated (1,000m^3)									
	Top Soil	**Inert soft C&D Material(1)**	**Rock (2)**	**Inert hard C&D Materials(3)**	**Bituminous Materials**	**Non-inert C&D Materials(4)**	**Contaminated Soil**	**Excavated Sediment**	**Chemical Waste**	**Asbestos Containing Material (ACM)**
Site clearance	80.0	0	0	390.0	40.0	210.0	TBC(5)	Minimal	Few hundred litres per month	TBC(6)
Site formation	0	1460.0	160.0	0	0	0				
Construction of new buildings and infrastructures	0	0	0	212.9	0	53.2				

Notes:

[1] "Inert soft C&D material" includes, but not limited to excavated soil, etc.

[2] "Rock" includes all grade rock.

[3] "Inert hard C&D materials" mainly is Artificial Hard Material (AHM) which includes, but not limited to, broken concrete and granular materials, etc.

[4] "Non-inert C&D materials" includes, but not limited to, bamboo, timber, paper and plastic, etc.

[5] TBC: To be confirmed. As the majority of sites are managed by private owner and still in operation, undertaking the environmental SI at this EIA stage is not feasible. The proposed environmental SI should be carried out once the operation of the sites is terminated and the land is resumed to determine the extent of land contamination.

[6] Further investigation is required by the asbestos specialist during the construction stage.

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Site Clearance

11.4.1.8 Site clearance material will be generated mainly from the site clearance of existing terrain. These materials will mainly consist of top soil, non-inert C&D materials, inert hard C&D materials, and bituminous materials.

11.4.1.9 It is estimated that around 80,000m^3 of top soil, 390,000m^3 of inert hard C&D materials, 210,000m^3 of non-inert C&D materials and 40,000m^3 of bituminous materials will be generated during site clearance. The quantities of material generated during site clearance in shown in **Table 11.3**

11.4.1.10 The inert C&D materials will be reused on-site for site formation and no off-site disposal would be required The non-inert C&D materials generated would be disposed of at landfill.

11.4.1.11 A detailed breakdown of the waste generated during site clearance in each phase of the Project as well as a summary of the recommended outlets for the C&D materials is summarised in **Table 11.8.**

Table 11.3: Summary of C&D material volumes during site clearance

Top Soil (1000m^3)	Inert soft C&D Material (1000m^3) [1]	Rock (1000m^3) [2]	Inert hard C&D Materials (1000m^3) [3]	Bituminous Materials (1000m^3)	Non-inert C&D Material (1000m^3) [4]
80.0	0	0	390.0	40.0	210.0

Notes:

[1] "Inert soft C&D material" includes, but not limited to, excavated soil.

[2] "Rock" includes all grade rock.

[3] "Inert hard C&D materials" mainly is Artificial Hard Material (AHM) which includes, but not limited to, broken concrete and granular materials, etc.

[4] "Non-inert C&D material" includes, but not limited to, bamboo, timber, paper and plastic, etc.

11.4.1.12 With proper implementation of good construction site practice and mitigation measures, the on-site handling and reuse of site clearance waste would not cause adverse environmental impacts.

Site Formation

11.4.1.13 Site formation waste will be generated mainly from the excavation works during site formation within the Project site. The waste will mainly consist of inert soft C&D material and rocks. With proper implementation of good construction site practice and mitigation measures, the on-site handling and reuse of inert C&D materials would not cause adverse environmental impacts.

Table 11.4: Summary of C&D material volumes during site formation

Top Soil (1000m^3)	Inert soft C&D Material (1000m^3) [1]	Rock (1000m^3) [2]	Inert hard C&D Materials (1000m^3) [3]	Bituminous Materials (1000m^3)	Non-inert C&D Material (1000m^3) [4]
0	1460.0	160.0	0	0	0

Notes:

[1] "Inert soft C&D material" includes, but not limited to, excavated soil.

[2] "Rock" includes all grade rock.

[3] "Inert hard C&D materials" mainly is Artificial Hard Material (AHM) which includes, but not limited to, broken concrete and granular materials, etc.

[4] "Non-inert C&D material" includes, but not limited to, bamboo, timber, paper and plastic, etc.

Construction of New Buildings and Infrastructures

11.4.1.14 C&D materials, including concrete, bricks, wood, steel, metal, glass, plastics, packing materials and general refuse, would be generated from the construction of new buildings and infrastructures (e.g. roads, sewage pumping stations, reclaimed water service reservoir, nullah revitalisation, sewerage network installation etc). The inert portion of the C&D materials is referred to public fill and the non-inert portion is referred to construction waste and would be disposed of at landfill.

11.4.1.15 C&D materials can be minimised through careful planning during the detailed design stage and with good site practice during construction. This includes the use of non-timber formwork and temporary works and on-site sorting of the C&D materials for reuse and recycling as far as practicable.

11.4.1.16 The total estimated gross floor area (GFA) of the proposed development in the assessment area is approximately 2,663,000m^2.

11.4.1.17 In accordance with the "Reduction of Construction Waste Final Report, Hong Kong Polytechnic University (March 1993)", a C&D materials generation rate of 0.1m^3 per 1m^2 of GFA is adopted. It is estimated that approximately total 266,300m^3 of C&D materials would be generated from construction of the new buildings and infrastructures of proposed development.

Excavated Sediment

11.4.1.18 Several small watercourses and ponds will be filled during the construction phase. These works may result in a small amount of excavated sediment. The excavated sediment to be generated from the construction will consist of soil, small poldered rocks and river / pond sediment.

11.4.1.19 There currently no information regarding the quantity and quality of sediment within the ponds and watercourse of the area available. Metal-based algaecides including copper-based algaecides are not commonly used for fresh water fish farming in Hong Kong due to the high cost and toxicity to fish. Calcium oxide, which is cheaper, non-toxic to fish and a more readily available chemical, is more commonly used to control algae and suspended solids in fish ponds. It is therefore considered unlikely that the excavated sediment will be contaminated. The excavated sediment, however, is expected to have high water content and high organic content. The excavated sediment is proposed to be stabilised / solidified by mixing with cement so that the mixture is suitable for reuse on-site. The handling of this mud may cause odour nuisance and water quality impacts if not properly managed.

11.4.1.20 For the Yuen Long Nullah revitalisation/decking works along Kung Um Road and Kiu Hing Road (DP10), since the current Yuen Long Nullah has a concrete-paved base and all possible sediment had been removed at the time of its construction, the revitalisation/ decking works will only involve demolition of the concrete-paved base and provide soft and hard landscaping to provide a place for leisure for both existing and future residents. As such, excavation of marine sediment is not anticipated.

11.4.1.21 All excavated sediment generated should be collected and handled in compliance with the Waste Disposal Ordinance. Soil mixing or cement mixing work is suggested to improve the physical properties of the excavated sediment such that the grading and plasticity of the mixture will be suitable for reuse on-site as backfilling materials in the Project. Mitigation and control requirements for excavated sediment are detailed in **Section 11.5.1**. Provided that excavated sediment is handled in

accordance with these requirements, the potential waste management implications, including potential hazards, air and odour emissions, noise and wastewater discharge, associated with handling of excavated sediment during the construction phase of the Project are not expected.

Contaminated Soil

11.4.1.22 As the Project site is managed by private owner and is still in operation, undertaking the environmental site investigation (SI) at this EIA stage is not feasible. The proposed environmental SI should be carried out once the operation is terminated and the land is resumed so that a site re-appraisal can be carried out to determine the extent of land contamination. The exact quantity of contaminated soil could only be estimated after site re-appraisal and site investigations. The findings of the re-appraisal will be documented appropriately through a supplementary Contamination Assessment Plan that would be submitted to EPD for agreement prior to the commencement of SI works. A Contamination Assessment Report, a Remediation Action Plan and a Remediation Report will also be prepared and submitted to EPD for agreement prior to commencement of the development works on the sites.

11.4.1.23 Should contamination was identified after site investigations, the contaminated soil should be remediated and reused onsite as far as practicable, and landfills disposal should be adopted only as a last resort in accordance with Section 4.1 of EPD's *Practice Guideline for Investigation and Remediation of Contamination Land*. The mitigation and control requirements for contaminated soil have been detailed in **Section 11.5.1**. Provided that the handling of contaminated soil are in accordance with these requirements, the potential waste management implications, including potential hazards, air and odour emissions, noise and wastewater discharge, associated with handling of contaminated soil are not expected. No adverse environmental concern is anticipated during the construction phase of the Project.

Asbestos Containing Materials (ACM)

11.4.1.24 Asbestos was widely used in the construction industry prior to the early 1980's for fireproofing, thermal and electrical insulation as well as in sound absorption materials. However, asbestos is currently recognized as hazardous materials, due to its etiological effects on human respiratory system.

11.4.1.25 As the proposed project involves the demolition of buildings/structures that were built before 1980's, large amount of ACM may be present in the buildings of PDA areas. Thus, ACM which may be disturbed during the demolition activities, should be removed and disposed of in a proper manner prior to the demolition work, so as to avoid the release of harmful asbestos fibres to the environment and minimise potential hazard.

11.4.1.26 All ACM if confirmed to be present within the existing premises must be removed and disposed of in accordance with the *Air pollution Control Ordinance* and the *Waste Disposal Ordinance* prior to the refurbishment work. A Registered Asbestos Consultant and Registered Asbestos Laboratory shall be engaged to conduct investigation for the presence of ACM. An Asbestos Investigation Report, an Asbestos Abatement Plan (AAP) (if required) and a notification of commencement of asbestos abatement works shall be submitted to EPD at least 28 days before the asbestos abatement works commences. Also, the removal of ACMs should be carried out by a Registered Asbestos Contractor according to the approved AAP under the supervision of a Registered Asbestos Consultant. The asbestos waste generated shall be disposed of by a licensed waste collector in compliance with the Waste Disposal Ordinance.

Chemical Waste

11.4.1.27 Materials classified as chemical waste are listed in the Waste Disposal (Chemical Waste) (General) Regulation. The major chemical waste types arising from the construction sites may include the following:

- Scrap batteries;
- Spent hydraulic oils and waste fuel;
- Spent lubrication oils and cleaning fluids from mechanical machinery; and
- Spent solvent from equipment cleaning activities.

11.4.1.28 Chemical waste may pose the following environmental, health and safety hazards if not stored and disposed of appropriately:

- Toxic effects to workers;
- Adverse effects on water and land from spills; and
- Fire hazards.

11.4.1.29 It is difficult to quantify the amount of chemical waste as it will be highly dependent on the Contractor's on-site maintenance practice and the quantities of plant and vehicles utilized. Nevertheless, it is anticipated that the quantity of chemical waste, such as lubricating oil and solvent produced from plant maintenance will be small and in the order of few hundred litres per month.

11.4.1.30 Storage, handling, transport and disposal of chemical waste should be arranged in accordance with the Code of Practice on the Packaging, Labelling and Storage of Chemical Waste published by the EPD. Chemical waste should be collected by a licensed collector and to be disposed of at a licensed chemical waste treatment and disposal facility. Wherever possible, opportunities for the reuse and recycling of materials will be taken. Mitigation measures for chemical wastes are detailed in **Section 11.5.1**. Provided that the handling, storage and disposal of chemical wastes are in accordance with these requirements, adverse environmental impacts are not expected.

11.4.1.31 The estimated amount of chemical waste to be generated during construction phase is summarized in **Table 11.5**.

Table 11.5 Summary of chemical waste during construction phase

Waste type	Total amount generated
Scrap batteries	A few hundred kilograms per month
Spend hydraulic oil and waste fuel	A few hundred litres per month
Spent lubrication oil and cleaning fluids	
Spend solvent	

General Refuse

11.4.1.32 The construction workers will generate refuse comprising food wastes, waste paper, aluminium cans and plastic bottles during construction period.

11.4.1.33 The storage of general refuse may give rise to adverse environmental impacts, such as water quality, odour and visual impact; and in the form of windblown litter, if not properly managed. The construction site may also attract pests and vermin if the storage areas are not well maintained and cleaned regularly. In addition, disposal of

waste at sites other than the approved disposal facilities could also lead to similar adverse impacts at those sites.

11.4.1.34 The number of work force (clerical and workers) to be employed for the Project is not available at this stage, but is anticipated not to be over 1,000 staff. Based on the generation rate of 0.65kg/person/day, the total refuse generated per day would be less than 650kg/day. Therefore, it is estimated that 4,271 tonnes of general refuse would be generated during construction phase. The detailed breakdown of estimated amount of general refuse to be generated during construction phase is summarised below.

Table 11.6 Summary of general refuse during construction phase

Activities	Period	Total amount generated (kg/day)
Construction phase	Approx. 18 years	< 650

11.4.1.35 In order to minimise the final disposal quantities of general refuse, provisions of recycle bins for different types of recyclable waste should be provided together with a general refuse bin. Arrangements should be made with the recycling companies to collect the recycle waste as required. The Contractor should implement an education programme for workers relating to avoiding, reducing, reusing and recycling general waste. Participation in a local collection scheme should be considered by the Contractor to facilitate waste reduction.

11.4.1.36 Provided that the mitigation measures are adopted, the potential environmental impacts caused by the storage, handling transport and disposal of general refuse are expected to be minimal. It is recommended that general refuse should be collected on a daily basis for disposal. Mitigation measures to minimise potential environmental impacts are recommended in **Section 11.5.1**.

Transportation Arrangement for Waste Disposal during Construction Phase

11.4.1.37 Land transport should be used to deliver and dispose of the waste generated from YLS development to the designated disposal outlets. It is expected that a maximum of 250 vehicles per day will be required to transport the generated waste during the construction phase. The tentative transportation routings for the disposal of various types of wastes generated during the construction phase are listed below. No barging points or conveyor systems will be established in the Project area. In addition, as in normal site formation work, the site formation work front will serve as the temporary stockpile area of the fill material. The temporary stockpiles will be removed in regular basis and there is no designated area for stockpiling throughout the construction period. The transportation routings may change subject to the actual traffic conditions of the roads. Nevertheless, with the implementation of appropriate mitigation measures (e.g. using water-tight containers and covered trucks), no adverse environmental impacts are expected due to the transportation of waste.

Table 11.7 Tentative Transportation Routings for Waste Disposal During Construction Phase

Type of Waste	Disposal Outlet	Tentative Transportation Routing
Non-inert C&D Materials, General Refuse	WENT Landfill	Via road network within YLS development, Yuen Long Highway, Tuen Mun Road, Wong Chu Road, Lung Fu Road, Lung Mun Road, Lung Kwu Tan Road and Nim Wan Road

Type of Waste	Disposal Outlet	Tentative Transportation Routing
Chemical Waste	CWTC	Via road network within YLS development, Yuen Long Highway, Tuen Mun Road, Tsing Long Highway, Tsing Sha Highway and Tsing Yi Road

Construction Phase Waste Summary

11.4.1.38 A summary of the construction waste arising from the works area with recommendation for outlets during construction phase is presented in **Table 11.8**.

Table 11.8 Summary of waste arising from the works area with recommendation for outlets during construction phase

Activities	Waste Type	Total Amount Generated (1,000m^3)				Total Amount Reused (1,000m^3)				Total Amount Disposed (1,000m^3)				Recommended Outlets
		WP1	WP2	WP3	WP4	WP1	WP2	WP3	WP4	WP1	WP2	WP3	WP4	
Site clearance	Top Soil	3.8	36.1	27.1	17.0	3.8	36.1	27.1	17.0	0	0	0	0	Reuse within the site
	Non-Inert C&D Materials [4]	5.1	81.1	99.9	22.4	0	0	0	0	5.1	81.1	99.9	22.4	Disposal to landfill
	Inert hard C&D Materials [3]	12.3	156.4	165.0	54.4	12.3	156.4	165.0	54.4	0	0	0	0	Reuse within the site
	Bituminous Material	1.8	17.4	13.0	8.2	1.8	17.4	13.0	8.2	0	0	0	0	Reuse within the site
	Excavated Sediment	Minimal				To be further decided based on sediment quality and quantity								Stabilised and solidified for reuse within the site as much as possible
	Contaminated Soil	TBC [5]				TBC [5]				TBC [5]				TBC [5]
	ACM	TBC[6]				TBC[6]				TBC[6]				TBC[6]
Site formation	Inert soft C&D materials [1]	9.7	630.6	417.5	399.9	9.7	630.6	417.5	399.9	0	0	0	0	Reuse within the site
	Rock [2]	1.0	70.1	46.4	44.4	0	0	0	0	1.0	70.1	46.4	44.4	Reuse in quarry or other projects [9]
Construction of new buildings and infrastructures[7][8]	Inert C&D Materials [3]	13.0	84.0	89.7	26.3	13.0	84.0	89.7	26.3	0	0	0	0	Reuse within the site as much as possible
	Non-inert C&D Materials [4]	3.3	21.0	22.4	6.6	0	0	0	0	3.3	21.0	22.4	6.6	Disposal to landfill

Activities	Waste Type	Total Amount Generated ($1,000m^3$)				Total Amount Reused ($1,000m^3$)				Total Amount Disposed ($1,000m^3$)				Recommended Outlets
		WP1	WP2	WP3	WP4	WP1	WP2	WP3	WP4	WP1	WP2	WP3	WP4	
General construction activities	General refuse	< 650 kg/day				-				< 650 kg/day				General refuse: Disposal to landfill
	Paper, metals, plastics etc.													Paper, metals and plastics etc.; Collected by recycler
	Chemical waste	Few hundred cubic meter per month				Few hundred cubic meter per month				Few hundred cubic meter per month				Recycled by licensed facility as far as possible, the remaining was disposal of at Chemical Waste Treatment Centre (CWTC) in Tsing Yi

Notes:

[1] "Inert soft C&D material" includes, but not limited to, top soil, excavated soil, etc.

[2] "Rock" includes all grade rock.

[3] "Inert hard C&D materials" mainly is Artificial Hard Material (AHM) which includes, but not limited to, broken concrete and granular materials, etc.

[4] "Non-inert C&D Materials" includes, but not limited to, bamboo, timber, paper and plastic, etc.

[5] TBC: To be confirmed. As the majority of sites are managed by private owner and still in operation, undertaking the environmental SI at this EIA stage is not feasible. The proposed environmental SI should be carried out once the operation of the sites is terminated and the land is resumed to determine the extent of land contamination.

[6] Further investigation is required by the asbestos specialist during the construction stage.

[7] Reduction of Construction Waste Final Report, Hong Kong Polytechnic University (March 1993)

[8] Approximate ratio for (inert waste) : (non-inert waste) is 8:2 "Monitoring of Solid Waste in Hong Kong, 1997" by EPD

[9] The rock can be properly sorted and exported to quarry for beneficial reuse as rock fill in other projects. As agreed with PFC and as stated in the C&DMMP, the rock will be sent to the quarry or other concurrent projects, but not to fill reception facilities.

[10] "WP" stands for "Works Package". Therefore "WP1", for example, stands for "Works Package Stage 1 Works" as referred in **Table 3.5**.
Construction period for WP1: Year 2020 – Year 2029; Construction period for WP2: Year 2022 – Year 2033; Construction period for WP3: Year 2031 – Year 2038;
Construction period for WP4: Year 2032 – Year 2038.

11.4.2 Operation Phase

11.4.2.1 The following activities, including the DP and non-DP elements in the RODP (as mentioned in **Sections 1.5** and **1.6**), have been included in the waste management implication assessment for the operation phase:

- Trunk Roads (TSWW Interchange) (DP1);
- Primary Distributor Roads (TYST Interchange) (DP2);
- Two new Distributor Roads (Road D1 to Road D2) (DP3);
- Environmentally Friendly Transport Services (EFTS) (DP4);
- Slip roads at the TYST Interchange (DP5);
- Partly depressed road/ underpass located at TSWW Interchange and full enclosures at TYST Interchange (DP6);
- New YLS STW (Site OU3.3) (DP7);
- Two new SPSs (Sites OU1.12 and OU2.2) (DP8);
- Reclaimed Water Service Reservoir for reuse of reclaimed water (DP9);
- Yuen Long Nullah revitalisation/decking along Kung Um Road and Kiu Hing Road (DP10);
- Residential Uses, Storage and Workshop Uses, Government, Institution or Community, Education, Public Utilities, Road and River Channels, Amenity, Open Space; and
- Other miscellaneous operations.

11.4.2.2 The waste implication during operation phase from the Project site has been assessed and presented in the following sub-sections.

11.4.2.3 The operation phase of the proposed development, including residential, educational and commercial establishments, in the Project site would generate the following categories of wastes based on their composition:

- municipal solid waste; and
- chemical waste.

11.4.2.4 The nature and quantity of each of these waste types arising from the operation of the proposed developments are described in the sub-sections below.

Municipal Solid Waste

11.4.2.5 Municipal Solid Waste (MSW) comprises of solid waste from households, commercial and industrial sources. With reference to the latest data from "*Monitoring of Solid Waste in Hong Kong 2015*" by EPD, the MSW disposal rate was 1.39 kg/person/day in Year 2015, and the recovery rate for recycling was 35% of the MSW generation. By calculation, the MSW generation rate, disposal rate and recycled rate were 2.14 kg/person/day, 1.39 kg/person/day and 0.75 kg/person/day in 2015 respectively. According to the "*Hong Kong Blueprint for Sustainable Use of Resources 2013 – 2022*" by EPD, a series of action agenda including waste reduction and pressure relief on landfills has been built upon enhanced social mobilization and implementation of policies and legislation. Necessary waste infrastructures to handle different types of waste will also be provided. The Blueprint has also set up ambitious

waste reduction targets which aim to reduce Hong Kong's MSW disposal rate on a per capita basis by 20%, from 1.27 kg per day to 1 kg or below by 2017, and then further down to 0.8 kg or below, representing a 40% from 2011. As such, the estimated quantities of MSW to be disposal of at the future development stage adopted for the purpose of EIA could serve as a conservative approach and would be reduced upon the achievement of the waste reduction targets as the programme progresses.

11.4.2.6 The estimated MSW based on planned residential and employment populations respectively is summarized in **Table 11.9**. The total estimated MSW generated from the entire YLS development is summarised in **Table 11.10**.

Table 11.9 Estimated quantities of MSW from Planned Residential and Employment Population during operation phase

Development Stage (Population Intake Year)	Estimated MSW from Residential Population (tpd) [1]				Estimated MSW from Employment Population (tpd) [1]			
	Residential Population	Generated [2]	Required Disposal [2]	Recycled [2]	Employment Population	Generated [2]	Required Disposal [2]	Recycled [2]
Stage 1 (2027)	9,860	21.1	13.7	7.4	180	0.4	0.3	0.1
Stage 2 (2033)	28,750	61.5	40.0	21.5	2,670	5.7	3.7	2.0
Stage 3 (2038)	38,750	82.9	53.9	29.0	4,600	9.8	6.4	3.4
Stage 4 (2038)	8,040	17.2	11.2	6.0	3,050	6.5	4.2	2.3
Total	**85,400**	**182.8**	**118.8**	**64.0**	**10,500**	**22.5**	**14.6**	**7.9**

Note:
[1] tpd: tonne per day.
[2] MSW disposal rate was **1.39 kg/person/day** and 65% of the MSW generation according to "Monitoring of Solid Waste in Hong Kong 2015" by EPD (https://www.wastereduction.gov.hk/sites/default/files/msw2015.pdf). By calculation, the MSW generation rate was **2.14 kg/person/day**. MSW recovery rate for recycling was 35% of the MSW generation according to "Monitoring of Solid Waste in Hong Kong 2015". By calculation, the MSW recycling rate was **0.75 kg/person/day**.

Table 11.10 Total Estimated quantities of MSW from YLS Development during operation phase

Development Stage (Population Intake Year)	Estimated MSW (tpd) [1]								
	Generated from			Required Disposal from			Recycled from		
	Residential	Employment	Total	Residential	Employment	Total	Residential	Employment	Total
Stage 1 (2027)	21.1	0.4	21.5	13.7	0.3	14.0	7.4	0.1	7.5
Stage 2 (2033)	61.5	5.7	67.2	40.0	3.7	43.7	21.5	2.0	23.5
Stage 3 (2038)	82.9	9.8	92.8	53.9	6.4	60.3	29.0	3.4	32.5
Stage 4 (2038)	17.2	6.5	23.7	11.2	4.2	15.4	6.0	2.3	8.3
Total	182.8	22.5	**205.2**	118.8	14.6	**133.4**	64.0	7.9	**71.8**

Note:
[1] tpd: tonne per day.

11.4.2.7 This estimate assumed no further waste reduction measure to reduce the demand for valuable landfill space. However, based on information from EPD, the major components of MSW in Hong Kong included glass, metals, paper, plastics and putrescible. Most of these materials are recyclable which could considerably further reduce the amount for final disposal.

Waste Collection and Disposal

11.4.2.8 An effective and efficient waste handling system is essential in order to minimize potential adverse environmental impacts during waste storage, collection and transport, such impacts may include odour if waste is not collected frequently; water quality if waste enters storm water drains; aesthetics and vermin problems if the waste storage area is not well maintained and cleaned regularly. The waste handling system may also facilitate materials recovery and recycling.

11.4.2.9 A refuse collection room would be installed in each building at the ground floor for localized refuse collection and the waste would be transported to a refuse collection point (RCP). Waste recycling facilities / containers are recommended to be included in RCP. The waste could be sorted to recover materials (such as paper and cardboards, plastics, metals and glass etc.) as far as possible, before to be compacted into mobile waste container. Different containers should be provided for the storage of different recyclable materials. To avoid potential odour nuisance to the residents during transport of waste, enclosed waste collection trucks should be used and the collection route and time should be properly planned. The RCP should contain mobile compactor and related equipment to provide adequate waste handling services. At least daily collection should be arranged by the waste collector.

Waste Recycling

11.4.2.10 In order to facilitate recycling, a 4-bin recycling system for paper, metals, plastics and glass should be adopted together with a general refuse bin. They should be placed in prominent places to promote waste separation at source. All recyclable materials should be collected by recyclers.

Chemical Waste

11.4.2.11 Paints, lubricants and used batteries may be generated during maintenance activities on the road networks within the YLS development. Moreover, educational institutions are planned within the proposed development and it is expected that chemical waste would also be produced from the laboratories of these educational institutions during the operation phase.

11.4.2.12 It is anticipated that the total quantity of chemical waste such as acids, alkalis and organic solvent produced by school laboratories would be insignificant.

11.4.2.13 To minimise potential environmental hazard due to waste handling, localized chemical waste storage areas should be located close to the source of waste generation for temporary storage. Drum-type containers with proper labelling should be used to collect chemical wastes for storage at the designated areas.

11.4.2.14 Registration as chemical waste producers with EPD should be made by representatives of the school prior to operation. All chemical wastes generated should be dealt with according to the Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes under the provisions of the Waste Disposal (Chemical Waste) (General) Regulation.

11.4.2.15 With the implementation of mitigation measures, potential environmental impacts (including potential hazard, air and odour emissions, noise, wastewater discharge and public transport) are not expected.

Screenings, Grits and Dewatered Sludge

11.4.2.16 The proposed YLS STW is located to the southern end of YLS PDA with design capacity of about 24,000m^3 per day designed to handle the sewage arising from the Project. Tertiary treatment and secondary plus treatment are assumed for the proposed YLS STW subject to detailed design. The major solid waste types produced from the operation are expected to be screenings, grits and dewatered sludge associated with the sewage treatment process. It is estimated that approximate 4.5m^3 per day of screenings and grits would be generated from the proposed YLS STW during the operation phase of the Project. In addition, assuming the sludge has gone through the thickening, anaerobic digestion and dewatering process within proposed YLS STW to reduce the volume as much as possible, it is estimated that around 14 m^3 per day of dewatered sludge would be generated during the operation phase of the Project. Since the proposed YLS STW is a separate Schedule 2 DP to be managed by future operator, the exact amount of the abovementioned solid waste types would be further quantified under separate EIA.

11.4.2.17 Nevertheless, as a recommendation, the generated screenings and grits shall be properly stored in a covered container and disposed of daily to the WENT Landfill. The transportation and disposal of the screenings and grits shall be managed and controlled by a reputable waste collector employed by the operator to reduce any potential pest odour and litter impacts. Dewatered sludge shall be properly stored in a covered container after the sludge dewatering process and disposed of daily to the existing sewage treatment facilities (STF) (e.g. STF in Tuen Mun). The dewatered sewage sludge shall be delivered by land transport in water tight containers or skips to avoid odour emission during transportation. Mitigation and control requirements for screenings, grits and dewatered sewage sludge are detailed in **Section 11.5.2**.

11.4.2.18 Provided that the handling, storage and disposal of screenings, grits and dewatered sludge are in accordance with these requirements, the potential waste management implications, including potential hazards, air and odour emissions, noise and wastewater discharge, associated with handling, storage and disposal of screenings, grits and dewatered sludge are not expected. No adverse environmental concern is anticipated during the operation phase of the Project.

Transportation Arrangement for Waste Disposal during Operation Phase

11.4.2.19 Land transport should be used to deliver and dispose of the waste generated from YLS development to the designated disposal outlets. It is expected that a maximum of 30 vehicles per day will be required to transport the generated waste during the operation phase. The tentative transportation routings for the disposal of various types of wastes generated during the operation phase are listed below. The transportation routings may change subject to the actual traffic conditions of the roads. Nevertheless, with the implementation of appropriate mitigation measures (e.g. using water-tight containers and covered trucks), no adverse environmental impacts are expected due to the transportation of waste.

Table 11.11 Tentative Transportation Routings for Waste Disposal During Operation Phase

Type of Waste	Disposal Outlet	Tentative Transportation Routing
MSW, Screenings and Grits	WENT Landfill	Via road network within YLS development, Yuen Long Highway, Tuen Mun Road, Wong Chu

Type of Waste	Disposal Outlet	Tentative Transportation Routing
		Road, Lung Fu Road, Lung Mun Road, Lung Kwu Tan Road and Nim Wan Road
Chemical Waste	CWTC	Via road network within YLS development, Yuen Long Highway, Tuen Mun Road, Tsing Long Highway, Tsing Sha Highway and Tsing Yi Road
STF	Dewatered Sludge	Via road network within YLS development, Yuen Long Highway, Tuen Mun Road, Wong Chu Road, Lung Fu Road, Lung Mun Road, Lung Kwu Tan Road and Nim Wan Road

Operation Phase Waste Summary

11.4.2.20 A summary of the operation waste arising from the works area with recommendation for outlets during operation phase is presented in **Table 11.12**.

Table 11.12 Summary of waste arising from the works area with recommendation for outlets during operation phase

Activities	Waste Type	Total Amount Generated	Total Amount Disposed	Total Amount Recycled	Recommended Outlets
Domestic waste generated from future residences of public and private housing	MSW	205.2 tonnes per day	133.4 tonnes per day	71.8 tonnes per day	Recycled at recycling facilities and/or disposed of at the proposed RCPs and/or the WENT Landfill
Commercial and industrial (C&I) waste generated from employees of enterprise and technology parks, offices and recreation sites					
Laboratory works within the educational institutions	Chemical waste	A few cubic metres per month	A few cubic metres per month	A few cubic metres per month	Recycled by licensed facilities or disposed of at the CWTC
Maintenance activities on the road networks within the YLS development					
Sewage treatment process within proposed YLS STW	Screening, grits and dewatered sludge	Screening and grits: Approximate 4.5m^3 per day Dewatered sludge: Approximate 14m^3 per day	Approximately 18.5m^3 per day	-	Screenings and grits are disposed of at the WENT Landfill Dewatered sludge are disposed of at the STF

11.5 Mitigation Measures

11.5.1 Construction Phase

11.5.1.1 The mitigation measures for construction phase are recommended based on the waste management hierarchy principles. Recommendations of good site practices, waste reduction measures as well as the waste transportation, storage and collection are described in following sub-sections.

Good Site Practice

11.5.1.2 Adverse waste management implications are not expected, provided that good site practices are strictly implemented. The following good site practices are recommended throughout the construction activities:

- nomination of an approved personnel, such as a site manager, to be responsible for the implementation of good site practices, arrangements for collection and effective disposal to an appropriate facility, of all wastes generated at the site;
- training of site personnel in site cleanliness, appropriate waste management procedures and concepts of waste reduction, reuse and recycling;
- provision of sufficient waste disposal points and regular collection for disposal;
- appropriate measures to minimise windblown litter and dust during transportation of waste by either covering trucks or by transporting wastes in enclosed containers;
- regular cleaning and maintenance programme for drainage systems, sumps and oil interceptors;
- a Waste Management Plan (WMP) should be prepared by the contractor and submitted to the Engineer for approval.

Waste Reduction Measures

11.5.1.3 Amount of waste generation can be significant reduced through good management and control. Waste reduction is best achieved at the planning and design phase, as well as by ensuring the implementation of good site practices. The following recommendations are proposed to achieve reduction:

- segregate and store different types of waste in different containers, skip or stockpiles to enhance reuse or recycling of materials and their proper disposal;
- proper storage and site practices to minimise the potential for damage and contamination of construction materials;
- plan and stock construction materials carefully to minimise amount of waste generated and avoid unnecessary generation of waste;
- sort out demolition debris and excavated materials from demolition works to recover reusable/recyclable portions (i.e. soil, broken concrete, metal etc.); and
- provide training to workers on the importance of appropriate waste management procedures, including waste reduction, reuse and recycling.

11.5.1.4 In addition to the above measures, specific mitigation measures are recommended for the specific waste types so as to minimise environmental impacts during handling, transportation and disposal of waste.

Storage, Collection and Transportation of Waste

11.5.1.5 Storage of waste on site may induce adverse environmental implications if not properly managed. The following recommendation should be implemented to minimise the impacts:

- waste such as soil should be handled and stored well to ensure secure containment;
- stockpiling area should be provided with covers and water spraying system to prevent materials from wind-blown or being washed away; and
- different locations should be designated to stockpile each material to enhance reuse.

11.5.1.6 The collection and transportation of waste from works area to respective disposal sites may also induce adverse environmental impacts if not properly managed. The following recommendation should be implemented to minimise the impacts:

- remove waste in timely manner;
- employ trucks and vessels with cover or enclosed containers for waste transportation;
- obtain relevant waste disposal permits from the appropriate authorities; and
- disposal of waste should be done at licensed waste disposal facilities.

11.5.1.7 In addition to the above measures, other specific mitigation measures on handling the excavated and C&D materials, chemical waste and materials generated from construction phase are recommended in the following subsections.

Excavated Sediment

11.5.1.8 The anticipated minor amount excavated sediment is proposed to be stabilised / solidified by mixing with cement so that the mixture is suitable to be reused on-site as backfill in construction of road base as much as possible. This approach avoids the need for off-site disposal that may result in impacts on the marine environment. Furthermore, all construction works near the watercourses should be carried out during dry season and within a dry zone to avoid adverse environmental impacts. The excavated sediment, if stockpiled on site, should be stored in enclosed containers and transported to the on-site treatment facilities as soon as practicable to minimise any potential odour impacts.

Site Formation and C&D Material

11.5.1.9 Wherever practicable, C&D materials should be segregated from other wastes to avoid contamination and ensure acceptability at Public Fill Reception Facilities areas or reclamation sites. The following mitigation measures should be implemented in handling the excavated and C&D materials:

- maintain temporary stockpiles and reuse excavated fill material for backfilling;
- carry out on-site sorting;

- make provisions in the Contract documents to allow and promote the use of recycled aggregates where appropriate; and
- implement a trip-ticket system for each works contract to ensure that the disposal of C&D materials are properly documented and verified.

Chemical Waste

11.5.1.10 For those processes which generated chemical waste, it may be possible to find alternatives to eliminate the use of chemicals, to reduce the generation quantities or to select a chemical type of less impact on environment, health and safety as far as possible.

11.5.1.11 If chemical wastes are produced at the construction site, the Contractors should register with EPD as chemical waste producers. Chemical wastes should be stored in appropriate containers and collected by a licensed chemical waste contractor. Chemical wastes (e.g. spent lubricant oil) should be recycled at an appropriate facility as far as possible, while the chemical waste that cannot be recycled should be disposed of at either the CWTC, or another licensed facility, in accordance with the Waste Disposal (Chemical Waste) (General) Regulation.

Contaminated Soil

11.5.1.12 It is considered unlikely that the issue on identified contaminated soil, if any subject to site investigation, would be a concern during either the construction or operation of the proposed development as appropriate remediation on the contaminated area would be carried out after land resumption prior to construction. Nevertheless, it is also recommended that implementation of good site practices, especially the procedures in handling the contaminated materials, should be implemented during the construction phase to minimise any potential exposure to contaminated soils or groundwater.

Asbestos Containing Materials

11.5.1.13 Due to the potential large amount of ACM during the site clearance stage, asbestos investigation is required. However, as asbestos investigation will involve a large number of buildings and most premises will involve private access, which cannot be obtained at this stage, it is considered that an asbestos specialist shall be employed by the responsible parties during the construction stage to investigate this issue.

11.5.1.14 Sufficient and reasonable lead time shall be allowed for preparation, vetting and implementation of asbestos investigation report and asbestos abatement plan in accordance with Air Pollution Control Ordinance, Cap. 311, before commencement of any demolition or site clearance work.

11.5.1.15 Some key precautionary measures related to the handling and disposal of asbestos based on Handling of Asbestos Containing Materials in Buildings (ProPECC PN 2/97) are listed as following:

- Adoption of protection, such as full containment, mini containment, or segregation of work area;
- Provision of decontamination facilities for cleaning of workings, equipment and bagged waste before leaving the work area;
- Adoption of engineering control techniques to prevent fibre release from work area, such as use of negative pressure equipment with high efficiency

particulate air (HEPA) filters to control air flow between the work area and the outside environment;

- Wetting of asbestos containing materials before and during disturbance, minimising the breakage and dropping of asbestos containing materials, and packing of debris and waste immediately after it is produced;
- Cleaning of work area by wet wiping and vacuuming with HEPA filtered vacuum cleaner;
- Coating on any surfaces previously in contact with or contained by asbestos with a sealant;
- Proper bagging, safe storage and disposal of asbestos and asbestos contaminated waste;
- Pre-treatment of all effluent from the work area before discharged; and
- Air monitoring strategy to check the leakage and clearance of the work area during and after the asbestos work.

General Refuse

11.5.1.16 General refuse should be stored in enclosed bins separately from construction and chemical wastes. Recycling bins should also be placed to encourage recycling. Preferably enclosed and covered areas should be provided for general refuse collection and routine cleaning for these areas should also be implemented to keep areas clean. A reputable waste collector should be employed to remove general refuse on a daily basis. It is expected that such arrangements would minimise potential environmental impacts.

11.5.2 Operation Phase

11.5.2.1 The following measures should be implemented on new developments to minimise the amount of waste to be disposed of at landfill and to maximize the recovery of material from the waste stream.

Municipal Solid Waste

11.5.2.2 General refuse from residential and commercial buildings should be collected with lidded bins and delivered to a central collection point and stored in enclosed containers to prevent windblown, vermin, water pollution and visual impact. At least daily collection should be arranged by the waste collector.

Chemical Waste

11.5.2.3 Paints, lubricants and used batteries may be generated during maintenance activities on the road networks within the YLS development. Furthermore, chemical wastes generated from the laboratories during operation would mainly include acid, alkali and solvent. To prevent health hazards to operators, all such chemical wastes should be collected and handled carefully.

11.5.2.4 To minimise potential environmental hazard due to waste handling, localized chemical waste storage areas should be located close to the source of waste generation for temporary storage. Drum-type containers with proper labelling should be used to collect chemical wastes for storage at the designated areas.

11.5.2.5 The producers should register with EPD as chemical waste producers. Chemical wastes should be stored in appropriate containers and collected by a licensed chemical waste contractor All chemical wastes generated from laboratories should be dealt with according to the Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes under the provisions of the Waste Disposal (Chemical Waste) (General) Regulation.

11.5.2.6 It is recommended that the chemical wastes are disposed at an appropriate waste disposal facility, such as the CWTC in Tsing Yi. A licensed contractor should be employed for the chemical waste collection.

Screenings, Grits and Sewage Sludge

11.5.2.7 The proposed YLS STW is designed to handle the sewage arising from the Project. The major solid waste types produced from the operation are expected to be screenings, grits and dewatered sludge associated with the sewage treatment process.

11.5.2.8 Although the proposed YLS STW is a separate Schedule 2 DP to be managed by future operator, it is still recommended to dispose of the generated screenings and grits to the WENT Landfill whereas the generated dewatered sludge shall be sent to existing STF (e.g. STF in Tuen Mun) for further treatment. The screenings, grits and dewatered sludge shall be delivered by land transport in water tight containers or skips to avoid odour emission during transportation. The solid waste unloading process shall be operated in the enclosed designated room inside YLS STW and served by negative pressure by extracting odorous gas to deodorising unit.

11.6 Residual Waste Management Implications

11.6.1.1 With the implementation of recommended mitigation measures for the handling, transportation and disposal of the identified waste, adverse residual waste management implications are not anticipated for both the construction and operation phases.

11.7 Conclusions and Recommendations

11.7.1 Construction Phase

11.7.1.1 Potential waste management implications from the generation of waste during the construction phase have been evaluated. Measures, including the opportunity for on-site sorting, reusing C&D materials etc., are devised in the construction methodology to minimise the surplus materials to be disposed. Recommendations have been made for implementation by the Contractor during the construction period to minimise waste generation and off-site disposal.

11.7.2 Operation Phase

11.7.2.1 The types of waste that would be generated during the operation phase have been assessed. Recommendations have been made to ensure proper treatment and disposal of these wastes.

11.8 References

- Hong Kong Polytechnic University (March 1993) Reduction of Construction Waste Final Report
- Approximated ratio for (inert waste) to (non-inert waste) is 8:2 (Monitoring of Solid Waste in Hong Kong, 1997)
- An Urgent Need to Suppress Waste Generation (EPD, 2010)